METRIC

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DETAIL SPECIFICATION

HFC-227ea FIRE EXTINGUISHING AGENT ENHANCED WITH SPECIAL SODIUM BICARBONATE POWDER (HFC-227ea/SBC_S)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and verification methods for two HFC-227ea/SBC_S Fire Extinguishing agents. For the purpose of simplicity, the term 'Agent' will be used to denote the HFC-227ea/SBC_S Fire Extinguishing agents. The purpose of these two Agents is to provide Army Aviation weapon systems, when in the proper hardware configuration, with an acceptable extinguisher configuration that does not employ an extinguishing mixture classified as an ozone-depleting substance (ODS).

1.2 Classification. The two Agents are free-flowing slurries of specialized sodium bicarbonate (SBC_S) powder suspended in HFC-227ea that will be used in fire extinguishing hardware (see 6.5 for a list of fire extinguishing hardware specifications) which is not covered in the scope of this document. The Agents are comprised of 1,1,1,2,3,3,3-Heptafluoropropane compound (generically named HFC-227ea) containing either of two SBC_S powders held in long term suspension as free-flowing slurries. The Agents will be further designated as either a Class I or Class II composition depending on the specific sodium bicarbonate employed in creating the slurry. The SBC_S powders employed in these Agents will be either of two types, designated as SBC-1 and SBC-2. The two Agent Classes are described in detail below and will be specified in the contract or purchase order (see 6.2).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Materials and Manufacturing Technology Branch, Specifications and Standards Office, Attn: RDRL-WMM-D, Aberdeen Proving Ground, MD 21005-5069 or emailed to richard.j.squillacioti.civ@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2.1 Class I. Class I Agents will contain SBC-1 (the sodium bicarbonate component prepared by size reduction processing (for example: utilizing Jet-Milling, as detailed in Appendix B) of United States Pharmacopeial (USP) grade sodium bicarbonate raw material).

1.2.2 Class II. Class II Agents will contain SBC-2 (from methods of processing USP grade sodium bicarbonate starting material that produce a preponderance of nanometer particles).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated (see 6.2), the issue in effect on date of invitation for bids or request for proposal should apply.

ASTM INTERNATIONAL

ASTM D6064 - Standard Specification for HFC-227ea, 1,1,1,2,3,3,3-Heptafluoropropane (CF3CHFCF3)

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order (see 6.2 and 6.4), a sample shall be subjected to first article inspection in accordance with (IAW) paragraph 4.2.

3.2 Agent sample source. Blended slurry samples shall not be obtained from a bulk tank that is pressurized above the vapor pressure of HFC-227ea. The only pressure within the bulk slurry tank shall be the vapor pressure of HFC-227ea that correlates to the temperature of the working environment.

3.3 Material and components. The materials and components shall conform to applicable specifications, standards, and patterns as specified herein and Table I. All the materials shall be capable of meeting all the operational and environmental requirements specified herein.

3.3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. HFC-227ea materials shall meet the requirements specified in ASTM D6064. Refer to paragraph 6.6.

3.3.2 Pre-Slurry and Post-Slurry Powder Requirements. The SBC_S employed in the preparation of the HFC-227ea/SBC_S suspensions shall be either SBC-1 or SBC-2 as specified in the contract or order (see paragraph 6.2). The resulting Agent slurries will be designated either Class I or Class II Agent, respectively. Materials shall not be added to improve the fluidity, appearance, and non-caking characteristics of the SBC_S. Note: Only USP grade sodium bicarbonate and materials derived by physical processing are acceptable. No chemical additives to the sodium bicarbonate or HFC-227ea are allowed.

3.3.3 Agent chemical and physical requirements. The SBC_S (Class I or Class II) moisture content, color, aspect ratio, surface area, size requirement and size distribution, shall be as specified in Table I and verified in accordance with 4.5.1, 4.5.2, 4.5.5, 4.5.6, 4.5.7, and 4.5.8. The agent's caking and lumping tendency, flow, suspendability and concentration shall be as specified in Table I and verified in accordance with 4.5.1, 4.5.3, 4.5.4 and 4.5.9.

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TABLE I. Agent Chemical and Physical Requirements

Requirement	Class I Agent (contains SBC-1)		Class II Agent (contains SBC-2)	
	Value	**Verification Paragraph**	**Value**	**Verification Paragraph**
Moisture Content of filtered slurry, maximum ppm	< 525 ppm	4.5.1 & 4.5.2	< 2000 ppm	4.5.1 & 4.5.2
Caking and Lumping tendency and Flow	None visible, Flows on rapid 90 degree tilting	4.5.3	None visible, Flows on rapid 90 degree tilting	4.5.3
Suspendability, in HFC-227ea	> 70% suspendability	4.5.4	> 95% suspendability	4.5.4
Color	Natural	4.5.1 & 4.5.5	Natural	4.5.1 & 4.5.5
Particle Aspect Ratio (width/length)	> 0.5	4.5.1 & 4.5.6	< 0.2	4.5.1 & 4.5.6
Particle surface area	between 12.8 and 15.6 m^2/g	4.5.1 & 4.5.7	between 19.4 and 23.6 m^2/g	4.5.1.& 4.5.7
Particle size requirement (longest dimension) (crystal lengths greater than 3 microns)	less than 1% of visible crystals greater than 3 µm in length	4.5.1 & 4.5.8	less than 1% of visible crystals greater than 3 µm in length	4.5.1 & 4.5.8
Particle size requirement (longest dimension) (crystal lengths greater than 1.5 microns)	no greater than 30% of visible crystals greater than 1.5 µm in length	4.5.1 & 4.5.8	no greater than 25% of visible crystals greater than 1.5 µm in length	4.5.1 & 4.5.8
SBC_S concentration in HFC-227ea in weight %	5.0 ± 0.5%	4.5.9	3.2 ± 0.2%	4.5.9

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the Agent. Blended slurry samples will be obtained from a bulk pressurized containers or from un-discharged fire extinguishers. This inspection shall include the examination of 4.4.3 and the tests of 4.5.1 through 4.5.8. Where needed, dry SBC can be extracted from blended slurry as described in 4.5.1.

4.3 Conformance inspection. Conformance inspection of the slurry sampled from the bulk slurry tank shall include the examination of 4.4.3 and the tests of 4.5.1 through 4.5.9.

4.3.1 Standard ambient conditions. Unless otherwise specified in the test procedures or in the contract or purchase order (see 6.2), all measurements and tests shall be made at standard ambient conditions. Standard ambient conditions shall be as follows:

Temperature:	$21^{o}C \pm 1.0^{o}C$ (70°F ± 1.8°F)
Relative Humidity (RH):	20 to 80%
Barometric Pressure:	Site pressure

4.4 Sampling.

4.4.1 Inspection lot. An inspection lot shall consist of five batches (see paragraph 6.3.5) or less when this fulfills the requirements of an order. When more than five batches are required to fill an order, the batches shall be grouped as equally as possible by weight to give not less than three and no more than five lots.

4.4.2 Submittal of product. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor shall submit the product in accordance with MIL-STD-1916.

4.4.3 Examination. Examination shall be performed for the requirements listed in TABLE I in accordance with MIL-STD-1916 with the specific acceptance level that is specified in the contract or purchase order (see 6.2). All nonconforming material shall be rejected.

4.5 Test methods. Each test shall be run in triplicate and the results averaged.

4.5.1 Dry box test method. Removal of SBC_S (Class I or Class II) and HFC-227ea from the slurry shall utilize the dry box method or equivalent. The dry box method requires the use of dry box procedures maintaining a relative humidity of less than 1% and special handling and specialized equipment designed to preserve the particle characteristics and the sample's water content for analysis.

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4.5.1.1 Equipment. Special equipment needs –nano particle filter (Donaldson Tetratex® TX6614 (AX03-767), 50 nanometer membrane bonded to polyester substrate) and filter housing (SKC #225-1712) with compatible fittings. This filter housing is used to hold the Donaldson filter and collect the SBC_S. Special handling needs – a dry box for sample handling. A top loading balance with a capacity of 200 ± 0.1 grams is required for sample weighing.

4.5.1.2 Procedure. The method for isolating dry SBC_S from the supplied bulk slurry tank involves initial agitation to re-suspend all contained material and generate a homogenous slurry of all bulk slurry tank contents and materials transfer of a calculated amount (based on the weight percent SBC_S and test needs) to a clean sampling cylinder. Once the volume of slurry to be sampled has been determined this amount is transferred by pumping to the sampling cylinder (see section 4.5.4) and the sub-sampling process begins. For sampling of the SBC_S attach the valve and filter housing (filter installed) to the sampling cylinder containing the slurry, shake this assembly and invert such that the cylinder can drain liquid HFC-227ea through the filter housing. The dry SBC_S still contained in the filter housing, is transferred (dry box procedure) into an open, pre-weighed jar and allowed to further "dry" (lose HFC-227ea absorbed onto particle surfaces) while inside the dry box. Once HFC-227ea loss is complete, the jar lid is tightened, weighed, and the jar sealed for shipment to the laboratory to perform water analyses, Field Emission Scanning Electron Microscopy (FESEM), and/or particle size distribution analysis.

4.5.2 Moisture content. The moisture content shall be determined by Gas Chromatography as outlined in Appendix A: The Analysis of Water in Sodium Bicarbonate by Gas Chromatography.

4.5.3 Caking and lumping tendency and flow. Applied to samples prepared for suspendability analysis in item 4.5.4 (Suspendability of SBC_S). Tubes prepared for suspendability analysis as outlined in paragraph 4.5.4 will be shaken vigorously for 60 seconds and immediately examined by holding the pressure tube in a near horizontal orientation and allowing the contents to flow back and forth as the tube is repeatedly rocked back and forth. Any apparent lumps or clumps of material that fail to separate into suspended particles under one further cycle of 60 seconds of vigorous shaking of the tube shall be taken as evidence of caking or lumping of the SBC_S, and the lot shall be rejected. Flow is evaluated similarly. Rapidly tilt the pressure tube 90° from the vertical and observe the flow of the slurry. The slurry should flow smoothly with no evidence of caking. If lumps of caked SBC_S are noted at any point in observing the slurry, the material is rejected. If the material fails to flow it is rejected.

4.5.4 Suspendability of SBC_S in HFC-227ea. SBC_S suspendability and flow will be characterized using a sample obtained accordingly. Sampling from a bulk slurry tank is administered by first sub-sampling the bulk slurry tank and then using the sub-sample in the suspendability characterization as described below. Samples obtained by other means shall be tested for suspendability in the same manner.

4.5.4.1 Bulk tank sampling of the agent (slurry).

4.5.4.1.1 Equipment. Pressure gauge (readable range 0 to 100 psi) with fittings for attachment to vapor port of the bulk tank; tank stand allowing full inversion of the bulk tank; cleaned sampling cylinder with valve and fitting appropriate for holding vacuum or pressure to be used to retrieve and hold bulk tank sample(s); filter housing with fitting adapter for attachment to the

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sampling cylinder's liquid sampling port; filter for filter housing; scale capable of weighing to 5 kg or greater with weight readings to the nearest 0.1 gram. A Donaldson Tetratex® TX6614 (AX03-767), 50-nanometer membrane bonded to polyester substrate and SKC #225-1712 filter housing are used to filter the slurry in order to obtain SBC_S samples. Andrews glass pressure tube (3-oz pressure tube part no. 110108003, working pressure = 225 psi) is used for sample suspendability, lumping, caking, color and flow evaluations.

4.5.4.1.2 Bulk tank sampling procedure. If a sampling cylinder was filled and partially emptied to acquire SBC_S for the % SBC_S composition or water analysis, the remaining sampling cylinder contents will suffice for determining the suspendability of SBC_S in HFC-227ea. In this case, skip steps 2 to 7 and go to step 8. Otherwise, the procedure shall follow steps 1 through 10, below.

1. Preliminary steps: check bulk tank pressure - it should be less than 70 psi if the tank is at room temperature of 70°F. If over 70 psi at 70°F, vent the bulk tank by briefly opening the vapor valve. Recheck and repeat until pressure is 70 psi or less.

2. Attach sampling cylinder to a vacuum pump and reduce pressure to less than 10 TORR (0.2 psi) and close sampling cylinder valve, disconnect from vacuum pump.

3. Weight the evacuated clean sampling cylinder to the nearest 0.1 gram and record this data on the supplied log sheet.

4. Strap bulk slurry tank into tank stand; secure as appropriate, attach evacuated sampling cylinder to the liquid side valve (do NOT open any valves at this time), invert bulk tank rapidly to thoroughly mix contents and create a uniform suspension for sampling, return bulk tank to upright position and invert an additional 5 times. Then, immediately open the liquid side bulk tank valve to the sampling cylinder valve to withdraw a sample of the bulk tank contents.

5. Close the bulk tank liquid sampling valve and the sampling cylinder valve and disconnect the sampling cylinder from the bulk tank. Immediately weigh the sampling cylinder and record its weight to the nearest 0.1 gram.

6. Compare the weight of the sampling cylinder and sample to the recorded weight of the empty sampling cylinder. If the weight of the sample contained is greater than 500 grams continue with step 7; if weight of the sample is less than 500 grams transfer additional slurry until the sample weight of 500 grams is achieved before continuing with step 7.

7. Place new filter in filter holder and attach to sampling cylinder valve and weigh the sampling cylinder with attached filter housing to the nearest 0.1 gram

8. Shake the sampling cylinder for a period of not less than 30 seconds to assure uniform suspension of its contents.

9. Evacuate an Andrews Glass pressure tube and close its valve. Attach the Andrews Glass pressure tube (or equivalent) to the sampling cylinder and invert the assembly two times. Then, fill the pressure tube to no more than 80% of its height with slurry.

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10. Disconnect the glass pressure tube from the sampling cylinder and proceed to testing the contents of the glass pressure tube for suspendability.

4.5.4.1.3 Analyzing the glass pressure tube slurry sample for suspendability. A sample of the slurry gathered in the Andrews glass pressure tube can be analyzed for suspendability by following steps 1 through 4, below:

1. Place the 3-ounce Andrews's glass tube in the vertical position along the side of a ruler and observe the maximum height of the HFC-227ea. Record this height and the time the measurement was taken.

2. After 4 hours recheck the height of the HFC-227ea. If the value is within 0.1 inches precede with measurement of the height of the SBC_S suspension to the nearest 0.1 inches.

3. Calculate the slurry suspendability by dividing the value obtained in step 2 by the value obtained in step 1 and multiplying by 100.

4. If the value calculated in Step 3 is less than 95% for SBC-2 (Class II), the test failed. If the value is less than 70% for SBC-1 (Class I), the test failed.

4.5.5 Color. Extracted samples of the slurry shall be visually inspected for color as specified in Table I. Compare a sample of the SBC_S powder to food grade commercial baking soda. The two samples must be nearly indistinguishable in color. No particles of rust or metal flakes or other extraneous materials are permitted.

4.5.6 SBC_S Particle Aspect Ratio. Aspect ratio is determined from analysis of field emission scanning electron microscopy (FESEM) photos of the SBC_S sample to obtain the largest crystal dimensions (lengths, widths) from which aspect ratios (width/lengths) can be calculated.

4.5.6.1 Preliminary material characterization step to obtain photos for use in characterization.

4.5.6.1.1 Procedure. Prepare FESEM samples as follows:

For FESEM where a circular 1-cm stud is used. Under dry nitrogen, a portion of the SBC_S should be placed into a small jar having an opening Inner Diameter (ID) less than 1 cm and sealed tightly. Keep FESEM laboratory room at 68° F or lower if possible. Avoid exposing the SBC_S to high humidity and store all samples and preparation materials in desiccators or dry-boxes. Place a piece of carbon tape onto the stud. Open the small jar and place the stud on top of the jar, tape facing the jar opening.

Gently shake the jar/stud. Tap the stud to remove loose SBC_S powder and quickly place into the FESEM sample chamber. Repeat the above but this time place into the sputtering tool for very light coat of gold. Remove and place into the FESEM sample chamber. Note: During sputtering, keep the gold coat to a minimum otherwise decomposition may occur. During analysis, compare the gold coated material to the un-coated material. The uncoated material may only endure a few tens of seconds under focus before the FESEM breaks down the SBC_S into sodium carbonate. Also "charging" for the larger particles is likely. Once in focus, moving to

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fresh portions of the stud may allow for resolution of detail. The gold coated material should last longer under the beam, but will eventually breakdown as well.

An alternative approach to the preparation of SBC_S is to paint an FESEM stud with a very thin coating of conductive paint and let dry. Using a fine pipette, take a single drop of the dispersion and place it on the painted stud and quickly place into the FESEM sample chamber. The Isopropyl Alcohol (IPA) will evaporate and the particles should stick to the IPA-rewetted paint surface. Repeat the above but this time place into the sputtering tool for very light coat of gold. Remove and place into the FESEM sample chamber. Obtain at least 10 high magnification (20,000 times) non-overlapping photos of the SBC_S sample with the measurement scale shown on the photo. Note: low energy beam currents are needed to avoid decomposing the sample.

4.5.6.2 Evaluation of the FESEM photos of SBC_S. Equipment needed - needle tipped drafting caliper, steel ruler (inch or centimeter ruling). The procedure shall follow steps 1 through 9, below.

1. Blow up photos to full page size and print out on high contrast printer. Label the printed pages in sequence. Label all printed pages with the lot traceability information, source of sample, and date of characterization.

2. Examine each numbered photo to identify the larger and longer crystals and mark each identified crystal numerically in sequence (1, 2, 3, 4 etc). Note: only select crystals whose boundaries are visible or substantially visible.

3. Using the caliper and ruler measure the scale on each photo and record the length of the scale (in inches or mm) and the scale value (microns).

4. Using the caliper and ruler, obtain length and width information (inches or mm) for the identified larger and or longer crystals.

5. Record all crystal length and width information (in inches or mm).

6. Convert all crystal length and width information to microns using the scale measurement as follows: length = measured length (obtained in step 5) times the scale (usually microns) divided by the length of the scale (determined in step 3)

7. Calculate aspect ratios by dividing width by length values of step 6 for each crystal.

8. Review the data and determine morphology and number of crystals of length exceeding limits for maximum length.

9. Review the data and determine morphology based on aspect ratio (width/length): Needle like samples will exhibit aspect ratios less than 0.2. Granular particles will exhibit aspect ratios (width/length) greater than 0.5.

4.5.7 Specific surface area test procedure. The method requires use of a Nuclear Magnetic Resonance (NMR) based surface area analyzer. The commercially available Acorn Area - Nuclear Magnetic Resonance (NMR) based surface area analyzer using AreaQuant software

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version 0.8.3. Manufacturer - Xigo Nanotools, 116 Research Drive Bethlehem, PA 18015 has been successfully employed to characterize both the Class I and Class II agents. The phone number for Xigo Nanotools is (610) 849-5090. See web-page http://www.xigonanotools.com/ for links to additional information.

The Acorn Area measures the wetted surface area of nano particle dispersions using a proton NMR technique. This method relies on the observation that liquid on the particle surface has a relaxation time orders of magnitude smaller than bulk liquid (i.e., liquid far away from the particle surface). The relaxation time of the dispersion is comprised of the response from both bulk and surface liquid, shifted from the bulk liquid in proportion to the quantity of liquid on the particle surface. The method is independent of the particle size distribution, shape, or porosity. However, the calculation of an absolute particle surface area requires a value for ka, a parameter which characterizes the relaxation time shift per unit area for a given particle-liquid combination. A ka value of 0.0026g/m2/ms is to be used in the calculation of surface area.

The suspension fluid is anhydrous n-dodecane (Sigma-Aldrich). Suspensions are prepared at approximately 13 wt% by the simple process of adding a fixed volume of n-dodecane to a known weight of the dry SBC_S powder. The proto-suspensions are then dispersed using a micro-homogenizer. Once prepared, the samples are then diluted by addition of n-dodecane until the “viscosity” is such that the suspension is capable of being the pipette (approximately 0.3 mL) into 4 mm ID glass NMR tubes. At this final concentration, virtually no sedimentation should be observed after 24 hours storage.

A single 0.5 ml aliquot of sample is placed in a 120 mm in length NMR tube, capped and inserted into the Acorn Area. Prior to the actual sample measurement the Xigo Nanotools verification reference solution is used to ensure that the instrument is properly functioning and that the reference frequency is correct. The verification standard is an aqueous solution of a paramagnetic material that replicates the relaxation behavior of a concentrated particulate dispersion without the complications associated with concentrated dispersions, such as settling, flocculation, etc. The verification standard has a relaxation time of approximately 45ms. The surface area is measured using a T2 measurement obtained from a CPMG sequence with 625 echo cycles and $\tau = 0.5$ ms, averaging 8 scans with a recycle delay of 5000 ms. See instrument procedures and operational instruction manual for data entry and sample introduction procedure. Two separate aliquots are taken from each bulk suspension. Each sample is measured 5 times. The specific surface area is calculated from the average T2 value (10 measurements in all) for each sample.

Note that characterizing specific surface area by measuring T2 (spin-spin relaxation time) in 1H-NMR is a well-established, generic technique, not one that requires a specific piece of software or specialized hardware. An online tutorial entitled "Measuring the Specific Surface Area of Nanoparticle Suspensions using NMR” can be viewed at http://cnx.org/content/m34663/latest/ . Alternate instrument vendors with NMR capabilities could adapt their instruments and develop equivalent surface area determination methods if needed.

4.5.8 Particle Size Requirement. As the larger sized particles govern critical performance factors this characterization is focused on ensuring that all lots of SBC_S’s do not exceed the

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length, aspect ratio (a morphology indicator) of materials used in the development of the two SBC_S's employed as additives to HFC-227ea. A careful visual review of high magnification field effect scanning electron (FESEM) photos will suffice to identify the largest single crystals. Measurement of crystal maximum length and the width allows calculation of the aspect ratio and an estimate of percent populations of longer crystals. The task of reviewing the FESEM images, identifying the larger crystals, measuring crystal lengths and widths and calculating aspect ratios is described under section 4.5.6. Review of the collected data to identify the upper bound of crystal sizes is performed and the results compared to the limits in TABLE I. The photographs as described in 4.5.6 shall be used to determine the percentages of particles greater than 1.5 and 3 microns.

4.5.9 SBC_S concentration in HFC-227ea. For determining the weight percentage of SBC_S in the bulk tank the following procedures shall be performed. Preliminary steps shall include checking the bulk tank pressure to ensure it is less than 70 psi and that the tank is at room temperature of 70 °F. Equipment shall include a pressure gauge (readable range 0 to 100psi) with fittings for attachment to vapor port of the bulk tank, tank stand allowing full inversion of the bulk tank; cleaned sampling cylinder with valve and fitting appropriate for holding vacuum or pressure to be used to retrieve and hold bulk tank sample(s); filter housing (SKC #225-1712. with fitting adapter for attachment to the sampling cylinder's liquid sampling port; filter material for filter housing; scale capable of weighing to 5 kg or greater with weight readings to the nearest 0.1 gram. The procedures for verification are shown in steps 1 through 13, below.

1. Attach sampling cylinder to a vacuum pump and reduce pressure to less than 10 torr (0.2 psi) and close sampling cylinder valve, disconnect from vacuum pump.

2. Weigh the evacuated clean sampling cylinder to the nearest 1 gram and record this data as "Weight of empty sampling cylinder".

3. Mix the contents of the bulk slurry tank thoroughly.

4. Attach the evacuated sampling cylinder to the bulk tank liquid sampling valve or slurry circulation line if so equipped. Once the tank contents are well mixed, open the appropriate valves to transfer slurry to the sampling cylinder and fill the sampling cylinder with the slurry. Disconnect the sampling cylinder. Immediately weigh the sampling cylinder and record its weight to the nearest 1 gram. Record this weight as the "Weight Cylinder plus slurry".

5. Compare the weight of the sampling cylinder and sample to the recorded weight of the empty sampling cylinder. If the weight of the slurry sample contained is greater than 750 grams continue.

6. With the sampling cylinder in a full UPRIGHT position (valve on top) slowly vent the sampling cylinder for 24hours to remove the liquid HFC-227ea.

7. Dry the sampling cylinder exterior to remove all condensed water. Pump down to 10 torr.

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8. Weigh the dried and vented sampling cylinder to the nearest 1 gram. Record the weight of the sampling cylinder and SBC_S content to the nearest gram. Record this weight as "Cylinder plus SBC_S".

9. Subtracting the cylinder plus SBC_S weight (step 8) from the full cylinder value (step 4) yields the weight of HFC-227ea initially contained in the slurry. Record this value as "HFC-227ea weight".

10. Remove the cylinder opening plug/valve assembly from the sampling cylinder. Shake the sampling cylinder and tap the exterior to remove as much SBC_S from the sampling cylinder as possible. Record weight of the empty sampling cylinder.

11. Subtract the "Empty sampling cylinder weight" (determined in step 10) from the weight of "Sampling cylinder plus SBC_S" (determined in step 8) to determine the "Weight of SBC_S in slurry" contained in the sampling cylinder. Record this weight as the "Weight of SBC_S in slurry".

12. Divide the "Weight of SBC_S in slurry" (step 11) by the sum of the HFC-227ea weight plus the "Weight of SBC_S in slurry" and multiply by 100. This is the value for the "% by weight SBC_S" in the agent slurry.

13. Respective to which class of SBC_S is being evaluated, the following determinations shall be made. If the value for "% by weight SBC_S" calculated in 11 above is outside the range for SBC-2 the test is failed. If the value is outside the range for SBC-1 the test is failed. Record a pass or fail as appropriate.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This specification will produce a HFC-227ea/SBC_S Agent which is not classified as an ozone-depleting substance (ODS) that will be used in the Fire Extinguishers (see 6.5).

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class (see 1.2).
c. If issues of documents are different (see 2.3).
d. When first article is required (see 3.1 and 6.4).
e. Either SBC-1 or SBC-2 in the preparation of the HFC-227ea/SBC_S suspensions (see 3.3.2).
f. If measurements and tests are to be made at non-standard ambient conditions (see 4.3.1).
g. If submittal of product is different (see 4.4.2).
h. Specify the specific acceptance level (see 4.4.3).
i. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Anhydrous n-Dodecane. Refer to the CAS Number 112-40-3, Formula $CH_3(CH_2)_{10}CH_3$, Molecular Weight 170.33

6.3.2 Agent. Agents are classified as Class I or Class II and are free-flowing slurries of either of two SBC_S powders suspended in HFC-227ea that will be used as Agent in fire extinguishing hardware.

6.3.3 Slurry. Either SBC-1 or SBC-2 powder in liquid HFC-227ea as a free-flowing suspension.

6.3.4 Agent Class. The Agent is either a Class I or Class II Agent with Class I Agent being a slurry of HFC-227ea and SBC-1 and Class II Agent being a slurry of HFC-227ea and SBC-2.

6.3.5 Batch. An agent batch size can vary and will be defined by the amount that can be produced where the physical properties meet the requirements. An agent batch size will also depend on the amount required to fill an order while sufficient samples can be extracted from this amount for evaluation. Lastly, the maximum batch size is defined by the material handling capability required to maintain a homogenous agent mixture during sampling (for evaluation) and during the filling of smaller vessels. No more than 5 batches are allowed per lot. An order will consist of no less than 3 lots and no more than 5 lots. Batch sizes within a lot should be of equal weight.

Example 1: An order is received for 450 lbs of agent (10% of this amount will be used for evaluation for compliance with MIL-DTL-32412 while the remaining amount will be used to charge fire extinguishers). The company placing the order can only agitate and rotate a bulk slurry tank weighing 140 lbs (including tank weight of 40 lbs). The agent manufacturer/blender can only keep his particle size, moisture content, etc. within the requirements of the specification if he makes batches of 100 lbs or less. The order will include 3 lots contain 2 equal batches weighing 75 lbs. each.

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6.4 First article. It is suggested that first article testing be required for every new contract especially if the possible vendor is new (has not produced this product before) or has not produced this product within the last six (6) months.

6.5 Fire extinguishing hardware specifications. The following is a list of acceptable specifications for fire extinguishing hardware.

a) MIL-DTL-32403; entitled: FIRE EXTINGUISHER, EMPLOYING HFC-227ea AGENT ENHANCED WITH SPECIAL SODIUM BICARBONATE POWDER (HFC-227ea/SBC_S), PORTABLE, WITH BRACKET

6.6 HFC-227ea. A certificate of analysis is recommended from the manufacturer that the HFC-227ea ingredient used in the extinguishing agent meets the requirements of ASTM D6064.

6.7 Subject term (key word) listing.

Clean agent
Fire suppression
Halon replacement
Hand held fire extinguisher (HHFE)
Nanometer particle
Nuclear Magnetic Resonance (NMR)
Slurry

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APPENDIX A

THE ANALYSIS of WATER in SODIUM BICARBONATE by GAS CHROMATOGRAPHY (AP GLI-Army C101)

A.1 SCOPE

A.1.1 Scope. This procedure was prepared for US Army Aberdeen Proving Ground, Aberdeen, MD to assess the content of water in Sodium Bicarbonate. The method described in this procedure measures water by gas chromatography (GC) using direct injection and thermal conductivity detection (TCD). There are no specifications for water content of sodium bicarbonate. Sample results are reported as <1% or ≥1% based on comparison to a standard. Quantitative results are also reported for reference purposes. The upper range of this analysis may be extended through dilution. The instrument may be calibrated lower to accommodate samples with low moisture content.

A.1.2 Applicability. The requirements specified in this appendix are required. The purpose of this procedure is to describe a method for the determination of water (moisture) in Sodium Bicarbonate for US Army Aberdeen Proving Ground, Aberdeen, MD. The procedure defines the reagents and equipment required for executing the method, the steps for carrying out the analysis, the controls for insuring quality of the results, and the calculation and reporting requirements.

A.2 APPLICABLE DOCUMENTS:

The document number for Galbraith Laboratories analytical method procedure outlined below is:

AP GLI-ARMY-C101, Rev. No. 1	-	Method to Determine Amount of Water in Sodium Bicarbonate Powder

Galbraith Laboratories, Inc.
Mailing Address: P.O. Box 51610, Knoxville, TN 37950-1610
Physical Address: 2323 Sycamore Drive, Knoxville, TN 37921-1750
Toll Free Telephone: 1-877-449-8797
Telephone: 1-865-546-1335
Fax: 1-865-546-7209 Fax
e-mail: labinfo@galbraith.com

A.3 REQUIREMENTS

A.3.1 Test procedure. A portion of the sample is accurately weighed directly into a centrifuge tube and immediately covered with a volume of isopropyl alcohol (IPA). The vial is closed tightly, mixed well to produce a slurry of the sample in the IPA, and sonicated/shaken to extract moisture from the sodium bicarbonate into the IPA. The extraction solution is filtered and is transferred to a GC vial for analysis. The injector will

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aspirate a portion of the solution followed by injecting a controlled volume of it into a gas chromatograph. The extracted water from the sample is detected using a thermal conductivity detector (TCD), and the moisture content is determined by comparison of the sample peak area to the areas of external standards using linear regression. Sample results are reported based on comparison to a Quality Control (QC) standard prepared to contain 1% water (limit test). Actual sample concentration from the calibration curve is also provided for reference purposes.

A.3.2 Instruments and Materials. The equipment and materials listed below or its equivalent shall be required to perform test/procedure.

1. Analytical balance with sensitivity of 0.01 mg
2. Assorted Class A volumetric flasks and pipets
3. Hamilton, S.G.E., or other equivalent glass microliter syringes
4. GC Vials
5. 15 or 50 mL Centrifuge Tubes, Fisher, or equivalent
6. Nitrogen or Argon glove box
7. Syringes, Disposable Polyethylene, Becton-Dickinson or equivalent
8. Filters, Disposable Nylon, pore size = 0.2 µm, 30 mm
9. HP 6890, or equivalent, equipped with the following:
    - 9a. A temperature controlled inlet with split/splitless capabilities and with a deactivated glass liner and a glass wool plug.
    - 9b. A RT-U-Bond, or equivalent, capillary column, 15 m x 0.53 mm x 20 µm df, coating as stated above, either with or without a guard column
    - 9c. Software controlled data system such as HP ChemStation, Dionex Chromeleon, or equivalent
10. Reagents
    - 10a. Isopropyl Alcohol, American Chemical Society (ACS) reagent grade, $\geq$ 99.9%, or equivalent
    - 10b. Dried Isopropyl Alcohol – Dry IPA over regenerated molecular sieves for at least 72 hours prior to use. Use at least 15% volume ratio of molecular sieves to solvent.
    - 10c. High purity air, helium and hydrogen, or equivalent
    - 10d. Molecular Sieves, 3Å, Sigma Aldrich, or equivalent
    - 10e. Regenerated molecular sieves – Heat molecular sieves for at least 4 hours at 350ºC. Store the regenerated sieves in a closed container within a desiccator.
11. Reference Standards
    - 11a. Water, USP purified

A.3.3 Procedure. Note: Moisture is pervasive in the environment. It is present in air, on the surface of laboratory equipment and in reagent solvents, even if they marketed by the manufacturer as “dry.” The technician is required to use sound techniques to minimize exposure of blanks, standards, and samples to uncontrolled sources of water. Dry all

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laboratory equipment in an oven at 105ºC for at least 30 minutes prior to use. Handle samples and prepare solutions in a glove box.

A.3.3.1 Blank Preparation.

A.3.3.1.1 Prepare in the same way as the sample, omitting the addition of the sample material.

A.3.3.1.2 The blank is carried through all sample preparation and analysis steps, and is used to monitor the moisture level of reagents, supplies, and the instrument system.

A.3.3.2 Standards Preparation. Note: Alternate schemes of preparing the following solutions are permissible as long as the final concentration is the same. For reference mL refers to milliliter and uL refers to microliter in the following procedure text.

A.3.3.2.1 Primary Stock Solutions (prepare each of these solutions fresh).

A.3.3.2.1.1 Primary Stock #1. Fill a 10 mL volumetric flask halfway with dry IPA. Dispense 100 µL of water into the flask and fill the flask to the mark with dry IPA. This solution is 10,000 mg/L.

A.3.3.2.1.2 Primary Stock #2. Prepare as indicated under Primary Stock #1 using a new volume of water.

A.3.3.2.2 Calibration Standards. Transfer 50, 100, 250, 500 and 1000 µL aliquots of the primary stock solutions into separate GC vials. Add enough dry IPA to each of the first 5 vials to make the final volume 1.0 mL, cap each and invert several times to mix. Do not add any dry IPA to the 1000 µL vial. Cap it and use as-is. Also prepare a calibration blank using the dry IPA. These volumes correspond to the following calibration levels: 500, 1000, 2500, 5000 and 10000 µg/mL, respectively. The instrument may be calibrated lower depending on the water content of the samples.

A.3.3.2.3 Independent Check Standards (ICS). This solution is used to verify the calibration curve and is used as a benchmark for comparison to the unknown sample solution. Transfer 250 µL of Primary Stock #2 into a separate GC vial. Add 750 µL of dry IPA to a vial to make the final volume 1.0 mL. Cap it and invert several times to mix. This solution is 2500 µg/mL.

A.3.3.3 Sample Preparation.

A.3.3.3.1 Flush a centrifuge tube with gas from the glove box (nitrogen or argon) and cap.

A.3.3.3.2 Obtain the tare mass (empty weight) of the centrifuge tube and return to the glove box.

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A.3.3.3.3 Dispense sample into the vial. Note: the product is likely to be fluffy.

A.3.3.3.4 Recap the vial and obtain the gross mass. Calculate the mass of the sample by subtracting the tare mass from the gross mass.

A.3.3.3.5 Return the sample to the glove box. Dispense 4 mL of dry IPA into the tube for every gram of sample. For example, if 2 grams of sample are in the tube, add 8 mL of dry IPA to the vial. Round the volume of IPA to the nearest 0.01 mL.

A.3.3.3.6 Return the cap and shake the tube. If a solution or slurry of the sample is not produced, additional IPA must be added until a solution or slurry is produced. Record the volume of additional IPA added.

A.3.3.3.7 Once a solution or slurry is produced, sonicate the sample for 30 minutes.

A.3.3.3.8 Allow the sample to separate. Remove a portion of the clear supernatant using a syringe and filter the fluid into a GC vial. If desirable, the solution may be centrifuged prior to filtering.

A.3.3.3.9 Cap the vial. This solution is now ready for analysis.

A.3.3.4 Matrix Spike (Positive Control). Prepare this solution as indicated for the sample, except add the spike during step A.4.3 after returning the sample to the glove box but before adding the dry IPA. Make certain to account for the volume of IPA added for the spike in the final volume of dry IPA added to the sample. Spike the solution with 375 µL of primary stock #2. This solution is 3750 µg/mL. This concentration corresponds to 1.5% (wt/wt).

A.3.3.5 Instrument Set Up. Set up and operate the instrument according to the instrument manufacturer's instructions for a direct injection analysis.

GC Conditions – for reference only, not required settings

Inlet Settings		**Detector Settings**		**Oven Parameters**	
Mode	Splitless	Makeup Flow (mL/min)	12	Initial (ºC)	100
Temperature (ºC)	115	Temperature (ºC)	180	Initial Hold (min)	6
Split flow (mL/min)	50			Ramp (ºC/min)	20
Purge time (min)	0.5			Final Hold (min)	20
Column Flow (mL/min)	2.1				

Instrument settings may be adjusted according to manufacturer's recommendation or to improve method performance.

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APPENDIX A

A.4 ANALYSIS

A.4.1 System Suitability. Analyze the working calibration standards, the ICS, and the blank preparation. If the following system suitability acceptance criteria are met, then proceed with the sample analysis. Otherwise, correct any problems that caused the system suitability failure and proceed once the problem has been corrected and confirmed by the system suitability checks. The coefficient of determination, r^2, of a plot of the working calibration standards against concentration (in µg of solvent) is not less than (NLT) 0.995 for each analyte. The value of the ICS is within 10% of the theoretical value.

A.4.2 Sample Analysis. Build the analysis queue within the instrument software. Place the set of vials containing the samples into the tray in the auto sampler in the same order indicated within the sequence.

A.4.3 Quality Control. The matrix spike sample (positive control) must read higher than the ICS. After every 10 samples (or fewer) and at the end of the run, continuing calibration verification checks (CCV) are made. Use these results also as part of the System Suitability checks. For each CCV injection, the criterion for acceptance is +/-15% of the target value. The Relative Standard Deviation (RSD) for all system suitability standards in the sequence must be ≤10%.

A.5 CALCULATIONS

A.5.1 Calculate the limit test result by comparing the solution concentration for the ICS to the solution concentration for the unknown sample. Calculate the reference result according to the following:

% (wt/wt) = [(A-B) x C] / (g sample)

Where:

A is the concentration of water in the sample solution
B is the concentration of water in the blank solution
C is the volume of extraction dry IPA added to the sample
"g sample" is the net mass of sample taken for the analysis

A.6 REPORT

A.6.1 Sample readings below the ICS should be reported as <1%; Sample readings equal to or above the ICS are reported as ≥1%. Also report the actual concentration from the calibration curve for reference purposes.

A.7 REFERENCES

A.7.1 Operating Manuals, Volumes 1 – 3, HP 6890 Series Gas Chromatograph, Hewlett Packard Company, Revision 1, October 1996.

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A.7.2 Galbraith Laboratories Inc., Standard Operating Procedure, GC-100D – Analysis of Residual Solvents by Gas Chromatography Using Direct Injection and Flame Ionization Detection, Revision 10, June 2011.

A.7.3 Williams and Lawton, Drying of Organic Solvents: Quantitative Evaluation of the Efficiency of Several Desiccants, J.Org. Chem., 2010, 75(24) pp 8351-4.

A.8 ALTERNATIVE METHOD for ANALYSIS of WATER in SODIUM BICARBONATE

A.8.1 Should the GC method not be available, an alternate, more expensive and more complicated procedure is listed in the following:

Karl Fischer based water content determination method: Determination of Moisture in Sodium Bicarbonate, L. N. Gard, R. C. Butler, *Anal. Chem.*, 1954, 26 (8), pp 1367–1368

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APPENDIX B

PROCESSING of JET-MILLED SODIUM BICARBONATE and HFC-227ea into a SLURRY

B.1 SCOPE

B.1.1 Scope. This procedure was prepared for the US Army Aberdeen Proving Ground, to produce a slurry of the Class I Agent. A Class I Agent slurry is the mixture of HFC-227ea and SBC-1. Before the slurry is mixed, the SBC-1 needs to be jet-milled to the micron particle size and, in a separate process, mixed "blended" with the HFC-227ea.

B.1.2 Applicability. The requirements specified in this appendix are required to meet Class I specifications (Table I). However, while this size reduction method was developed for the purpose of this Appendix other equivalent size reduction processes may exist. The purpose of this procedure is to describe a method for producing a slurry of HFC-227ea and SBC-1 where the SBC-1 is jet-milled to the micron particle size and is then blended with the HFC-227ea to produce a slurry of SBC_S suspended in HFC-227ea that will be used in fire extinguishing hardware. The procedure defines the reagents and equipment required for executing the method, the steps for carrying out the analysis, the controls for insuring quality of the results, and the calculation and reporting requirements.

B.2 APPLICABLE DOCUMENTS: (Not Applicable)

B.3 REQUIREMENTS

B.3.1 Test procedure.

B.3.1.1 Jet-Milling of Sodium Bicarbonate to prepare SBC-1. Start with USP grade sodium bicarbonate. Table B-I contains the parameters used to make the jet-milled material SBC-1. Figure B-1 shows a photograph of the jet-milling equipment employed. Table B-II shows the difference in particle size distributions between the feed material and the processed material for one of the batches. The values given are weight by material volume. The particle size measurement instrumentation used was a Malvern Mastersizer 2000. The particle size values listed in Table B-II were obtained from the Malvern Mastersizer and are not required - they are provided for reference only.

B.3.1.2 Processing Equipment. Table B-III contains a list of the processing equipment used to process the jet-milled sodium bicarbonate and HFC-227ea into a slurry. The equipment is assembled using chemical resistant build-your-own hoses according to Figure B-2.

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APPENDIX B

TABLE B-I Parameters used to jet-mill the sodium bicarbonate. (for reference only)

PARAMETER	VALUE
Micronizer Size/Type	8” Open Manifold
Liner Type	Stainless Steel
Total Air Flow	130 Scfm @ 100 Psig, dry nitrogen
Grind Pressure	105 Psi
Feed Pressure	80 Psi
Venturi Throat Dia.	1/4” Dia. Al. Oxide
Feed Nozzle Dia.	# 30
Vortex Finder Dia.	2.0” Diameter
Feeder Type/Size	K-Tron Twin / 35 mm
Feeder Setting	Std. CC Screws / 70 - L
Feed Rate	2 lbs/hr.



FIGURE B-1. Photograph of jet-milling equipment.

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APPENDIX B

TABLE B-II. SBC-1 Particle size distribution (for reference only).

FEED	PROCESSED
d10 = 39.09µm	d10 = .665µm
d50 = 78.94µm	d50 = 1.421µm
d90 = 139.95µm	d90 = 2.805µm
d100 = 208.93µm	d100 = 5.754µm

TABLE B-III. Slurry Processing Equipment.

MANUFACTURER/SUPPLIER	SPECIFICATIONS
De Dietrich	20 gallon reactor, steel, glass lined
Corken	D91 vapor compressor
Q Sonica	700 Watt Sonicator
Q Sonica	400 ml high continuous flow flocell
Warren Rupp	S05 stainless steel diaphragm pump
Edwards	E2M2 vacuum pump
McMaster Carr	Flat-Panel Open-Tank Heat Exchanger, 316L stainless steel
Worthington Cylinders	239 lb refrigerant recovery cylinder
Mixer Direct	1/2 HP Flange Mount Gear Driven Variable Speed Mixer w/ 6” Propeller

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APPENDIX B

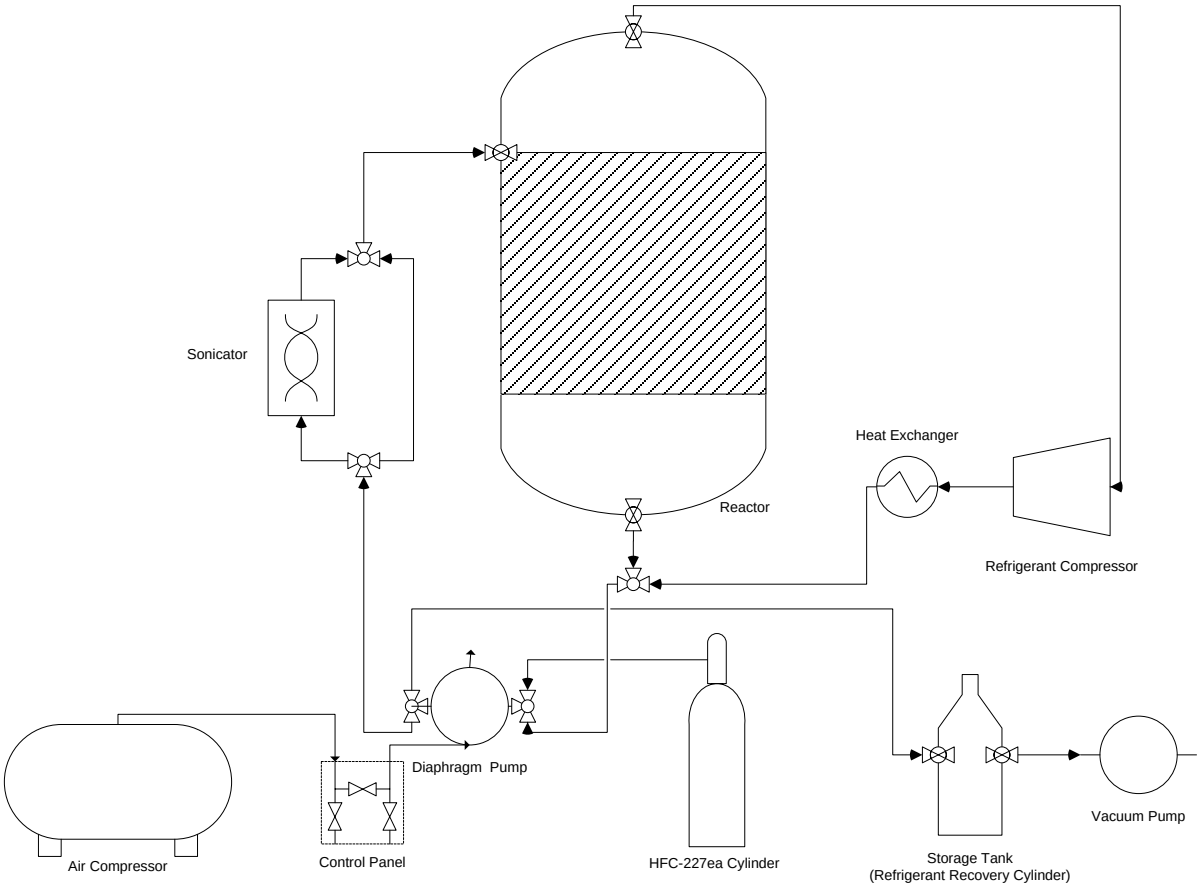


FIGURE B-2. Slurry processing setup.

B.3.1.3 Filling the Reactor with SBC-1 and HFC-227ea. Ensure all the reactor valves are closed prior to adding the SBC-1. Weigh 4276 grams of SBC-1 and transfer it into the Reactor through the view port. Close the view port and attach the clamps, such that, the view port is flush with the reactor and the vapor is not allowed to escape into the Reactor. Set the HFC-227ea cylinder onto the scale and tare the scale. Open the liquid valve on the HFC-227ea cylinder and the valve to the Reactor. Using the diaphragm pump, pump 179 pounds of HFC-227ea into the reactor. Turn off the diaphragm pump and close the reactor valve. A photograph of the Reactor is shown in Figure B-3.

B.3.1.4 Initial Break-Up of Sodium Bicarbonate at the Bottom of the Tank. Use the compressor to pump HFC-227ea vapor through the bottom of the reactor and break up the SBC-1 that has settled. Allow the compressor to pump through the bottom for one minute. As the vapor is being pumped through, turn on the stirrer and begin mixing the slurry. Turn off the compressor and close the valves while continuing to run the mixer. Figure B-4 shows a photograph of the compressor.

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APPENDIX B



FIGURE B-3. Photograph of Reactor.



FIGURE B-4. Photograph of the compressor.

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B.3.1.5 Sonication. Use the diaphragm pump to pump the slurry through the flocell. As the slurry flows through the flocell, turn on the sonicator system and begin sonication. Monitor the wattage and power on the sonication system throughout the process. The target wattage is 200 W. The sonicator should be cycled on and off to prevent overheating. Follow recommendations provided in the user manual. Sample the slurry using the sampling procedure and continue sonication until the slurry displays adequate suspension properties. FIGURE B-5 shows a photograph of the flocell and sonicating probe.



FIGURE B-5. Photograph of the flocell and sonicating probe.

B.3.1.6 Sampling the Slurry. Pull a vacuum on the Andrews glass pressure jar (Part number 110108 0006) and weigh the empty glass pressure jar (W_{empty}). Connect the glass pressure jar to the sample port on the reactor and fill the glass pressure jar with slurry. Close the glass pressure jar and disconnect from the sampling port. Weigh the sampling jar containing slurry (W_{Total}). Weight of the slurry (W_{slurry}) equals W_{Total} minus W_{empty}. Slowly vent the sampling jar by cracking open the ball valve. Be careful not to allow the SBC-1 to escape the sampling jar with the HFC-227ea vapor by controlling the flow through the ball valve. Once the HFC-227ea has been vented, pull a vacuum on the sampling jar. Again, ensure the SBC-1 does not escape the sampling jar with the HFC-227ea vapor by controlling

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the flow through the ball valve. Once the flow of HFC-227ea has stopped and a vacuum is on the sampling jar, close the sampling jar ball valve. Weigh the sampling jar containing the residual SBC-1 ($W_{Total-1}$). Weight of the SBC-1 (W_{SBC}) equals $W_{Total-1}$ minus W_{empty}. Use the following equation to solve for the mass percent of SBC-1 in the slurry (m_{SBC}).

$$m_{SBC} = 100\, X \frac{W_{SBC}}{W_{slurry}}$$

B.3.1.7 Transport to Storage Tank. Using a clean 239-lb refrigerant recovery cylinder (storage tank) under vacuum, connect the vapor recovery valve to the diaphragm pump and pump the slurry from the reactor into the storage tank. Continue pumping until the reactor is empty or the storage tank is at its rated capacity. If the storage tank reaches its rated capacity, isolate the line to the storage tank and connect a new tank and continue filling with slurry.

B.4 ANALYSIS

B.4.1 Correcting Mass Percent of SBC-1. The target mass percent of SBC-1 is 5% w/w. However, any losses of HFC-227ea vapor during processing will alter the mass percent of SBC-1. HFC-227ea may have to be added to correct the mass percent. The equation to calculate how much HFC-227ea to add is given below.

$$W_{HFC-227ea} = m_{SBC} W_{slurry} \left(\frac{1}{5.0} - \frac{1}{m_{SBC}} \right)$$

where, $W_{HFC-227ea}$ is the weight of HFC-227ea to add,
m_{SBC} is the mass percent of SBC-1 previously determined during the sampling step, and
W_{slurry} is the weight of the slurry in the storage tank.

Once the amount of additional HFC-227ea required to achieve a 5% w/w value has been determined, and this calculated amount of HFC-227ea transferred to the reactor, recirculate the reactor contents to ensure thorough mixing.

If the weight % of SBC is too low there is excess HFC-227ea in the slurry. Calculate the total excess weight using the above equation. Using a balance and a refrigerant recovery unit attached to the slurry tank vapor port remove HFC-227ea vapor and transfer it to the recovery tank in the following manner. Weigh the empty recovery tank as it sits on the balance and then start the recovery unit's pump. Stop the process once the weight increase of the recovery tank is equal to the amount of excess HFC-227ea calculated.

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CONCLUDING MATERIAL

Custodian:
Army - MR
Navy – AS
Air Force - 99

Preparing Activity:

Army - MR
(Project 4210-2012-004)

Review Activities:
Army – AT, AV, MI
Navy – YD
Air Force – 03, 11, 84
DLA – CC, GS3, IS
GSA - FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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